Backdoor criterion in action

Reminders from yesterday

- We revised the causal elementary blocks in DAGs:
  - fork, pipe, collider & descendant
- We saw that in some DAGs, blocking some paths could lead to open undesired other paths.
- We introduced the do-calculus and the backdoor criterion to decide if a direct effect between a treatment and an outcome was possible to estimate from a DAG perspective.

Backdoor criterion in action

Backdoor criterion is a rule to ding a set of variables to stratify by (condition on) to yield $P(Y|do(x))$

3 Steps:

- Identify all paths connecting treatment X to outcome Y
- Paths with arrows entering X are backdoor paths (non causal paths)
- Find adjustment set that closes/blocks all backdoor paths

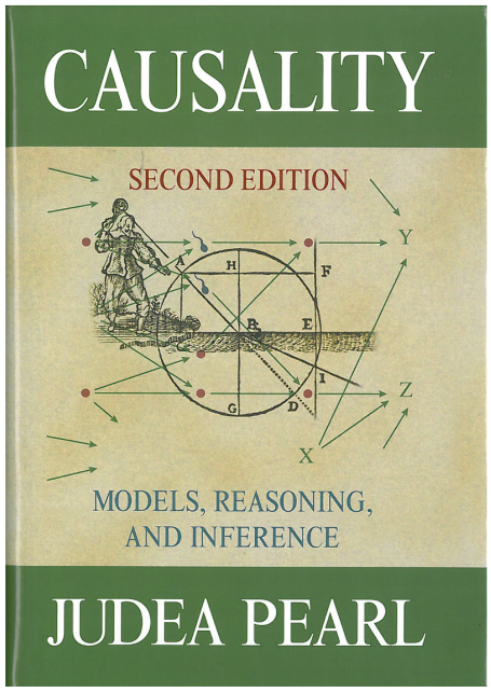
Where we are going

Goals for today:

- Identify backdoor paths
- Identify which paths are open/closed
- Close paths as necessary (stratify)

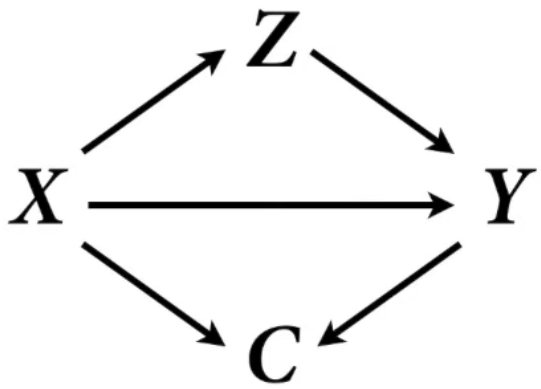
Shutting the back door

- The backdoor criterion: confounding by existence of an open back door paths from X to Y.
- If you know all the elements, you know how to open/close each of the paths. Back door or front door ones.



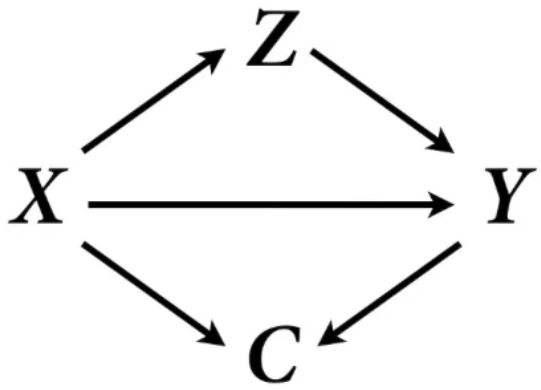
How many paths?

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.



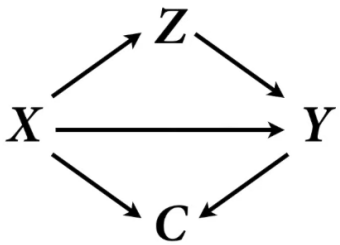
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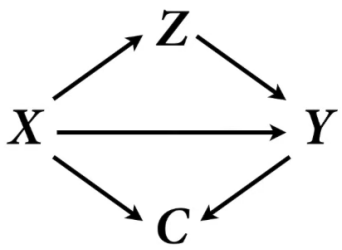
There are three paths in this case.

How many paths?



- Paths are $X \to Y$ (direct),
- $X \to Z \to Y$ (Z mediator) and
- $X \leftarrow C \to Y$ (C collider).

How many paths?



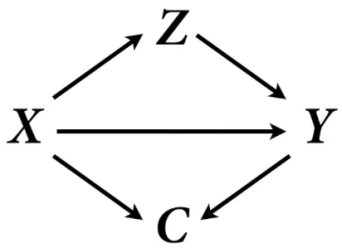
Can be done in R:

```
library(dagitty)
dag_XYZC <- dagitty("dag{
  X -> Z -> Y
  X -> Y
  X -> C <- Y}")
```

```
paths(dag_XYZC, from = "X", to = "Y")
```

```
## $paths
## [1] "X -> C <- Y" "X -> Y"      "X -> Z -> Y"
##
## $open
## [1] FALSE  TRUE  TRUE
```

How many paths?




```
library(dagitty)
dag_XYZC <- dagitty("dag{
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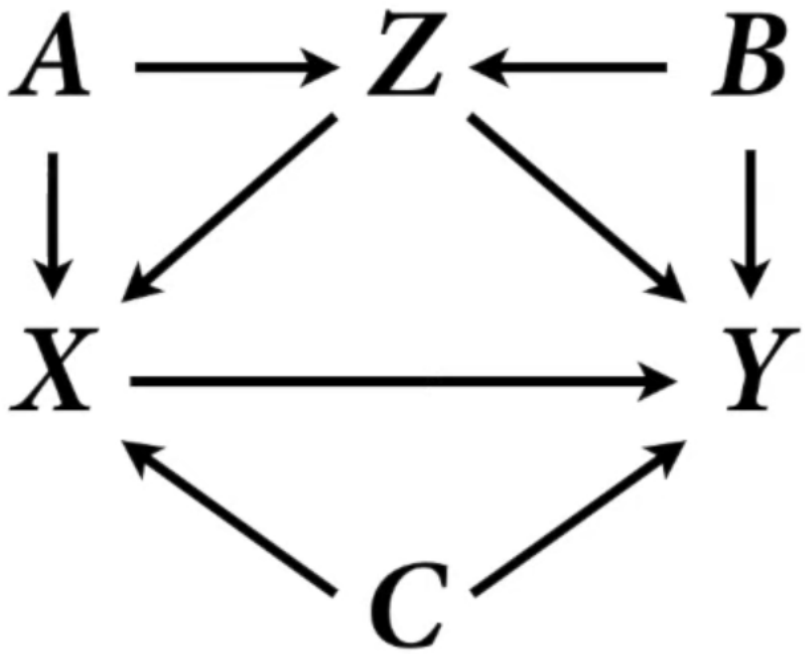
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```
## $paths
## [1] "X -> C <- Y" "X -> Y"      "X -> Z -> Y"
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## $open
## [1] FALSE  TRUE  TRUE
```

- The first (backdoor) is closed (because C is a collider); do not open it. The second one is open, we want to keep it open and it cannot be closed. The third path is a frontdoor path and is open.
- If we want the total effect of X on Y, we don't need to condition!

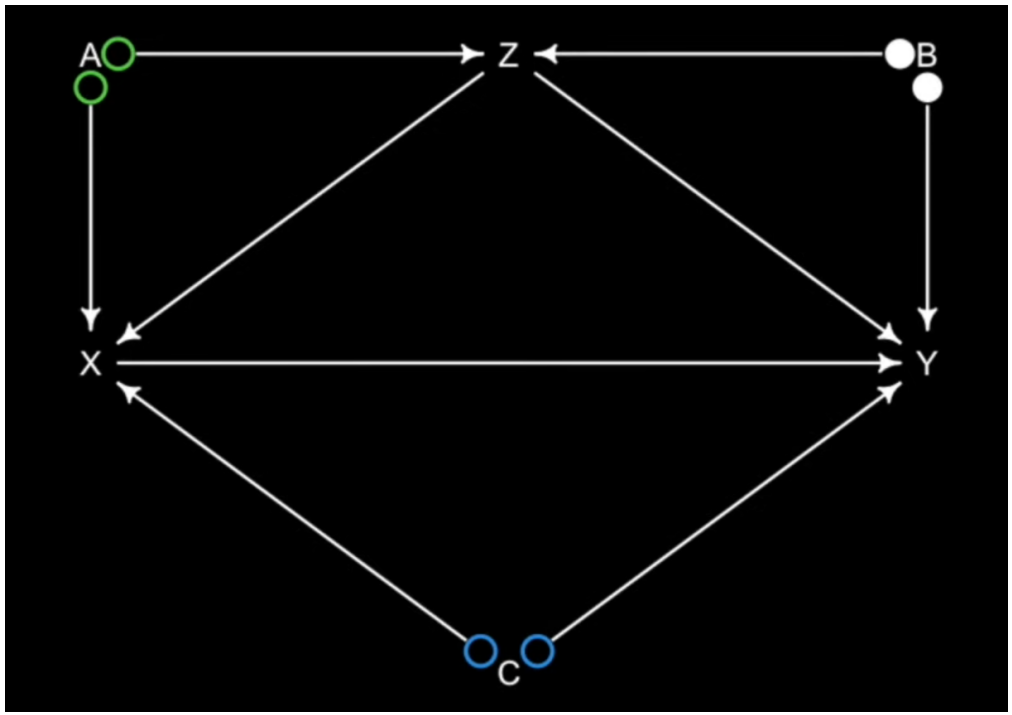
Level up

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.



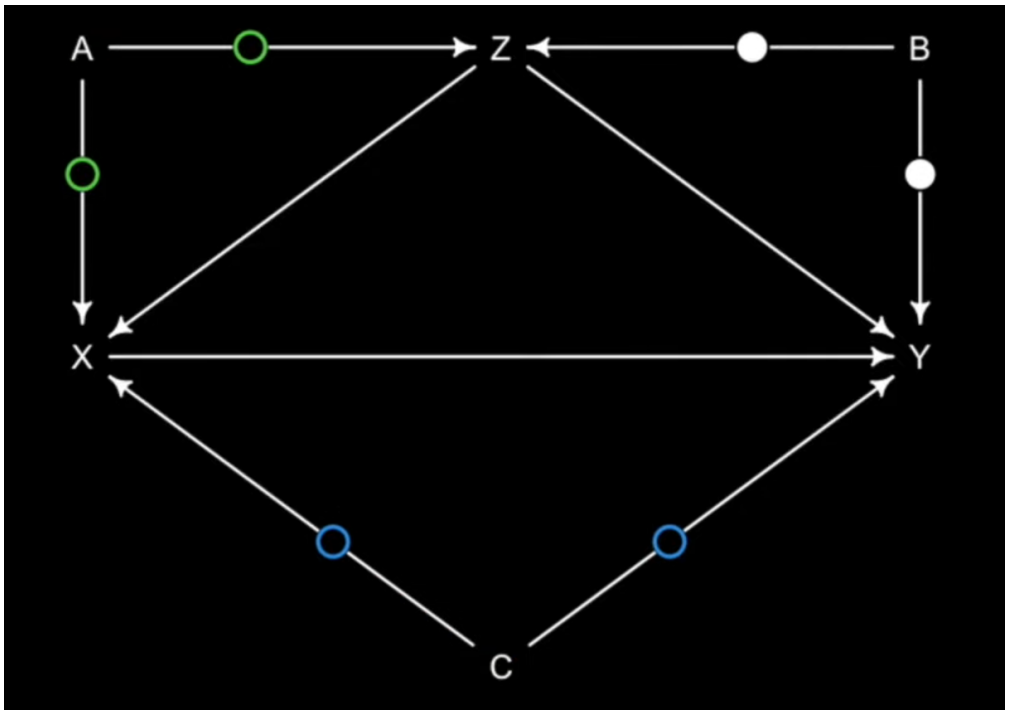
Flow of information

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.



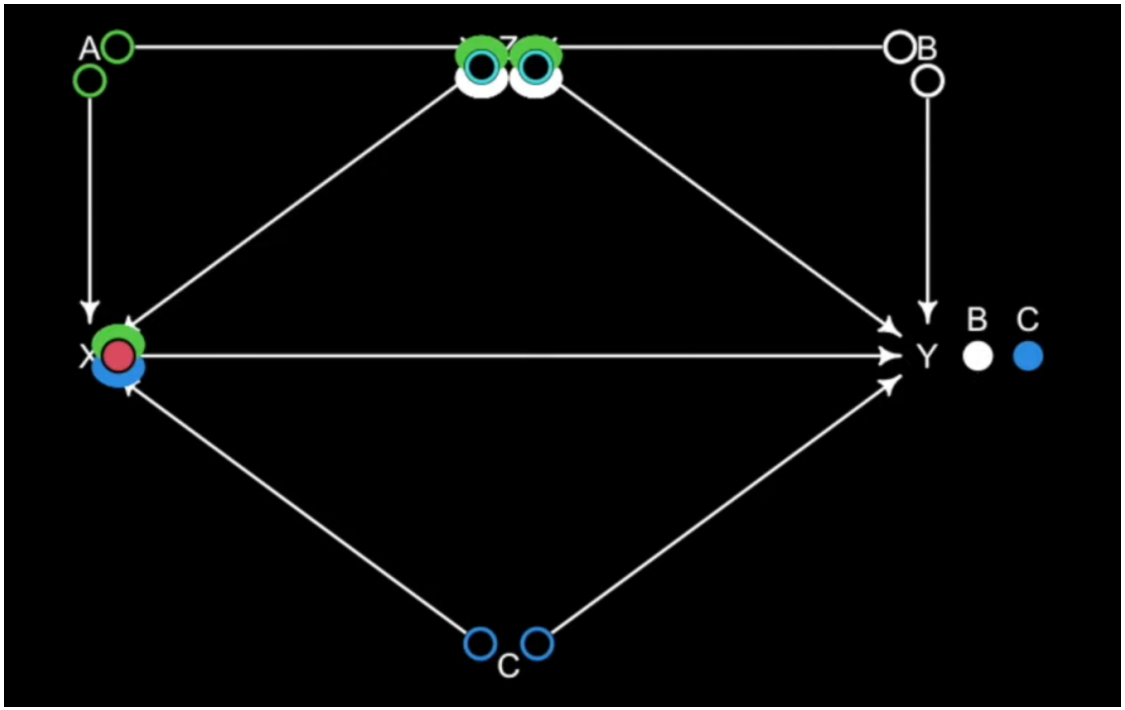
Another example: flow of information

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.



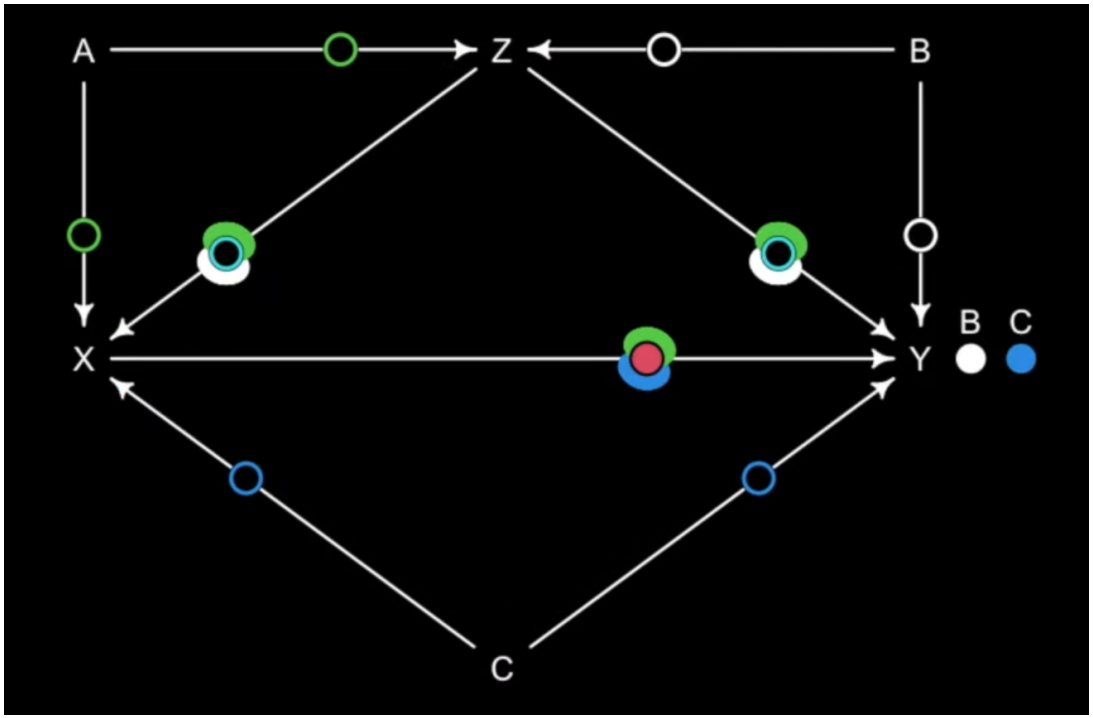
And another

List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.



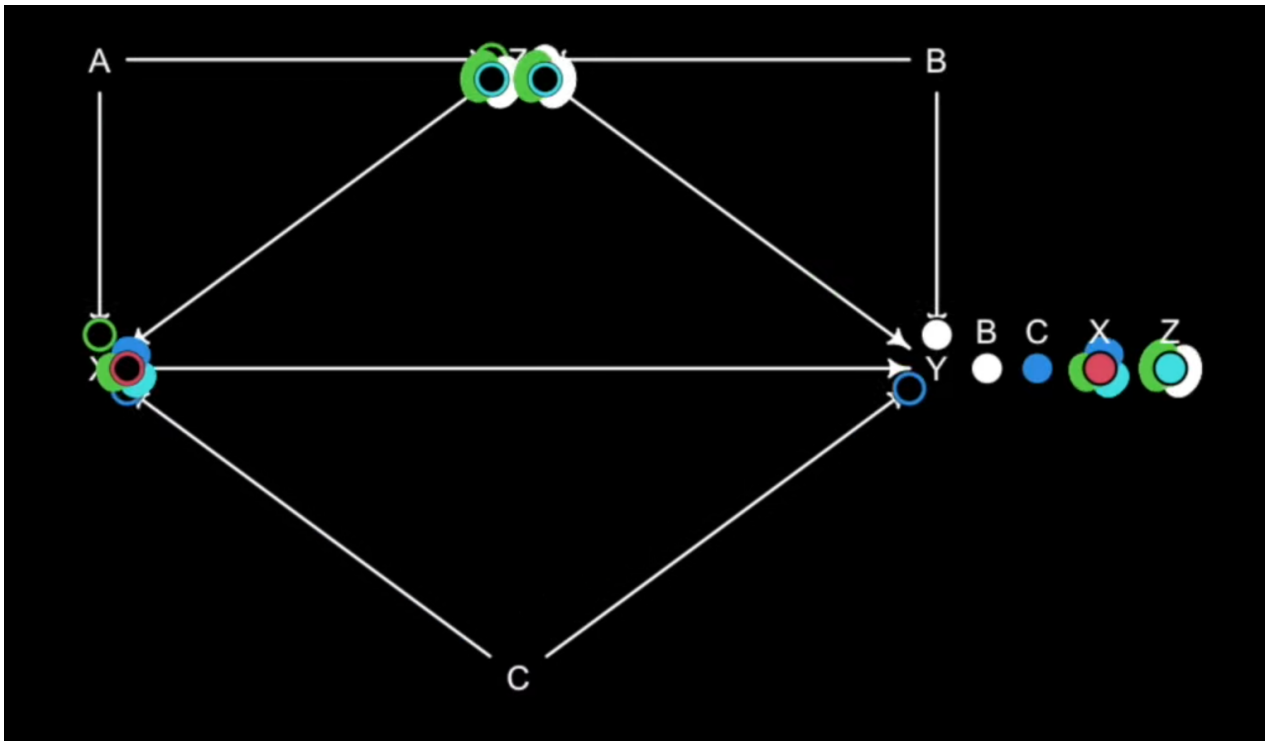
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List all the paths connecting X to Y, which need to be closed to estimate the effect of X on Y.

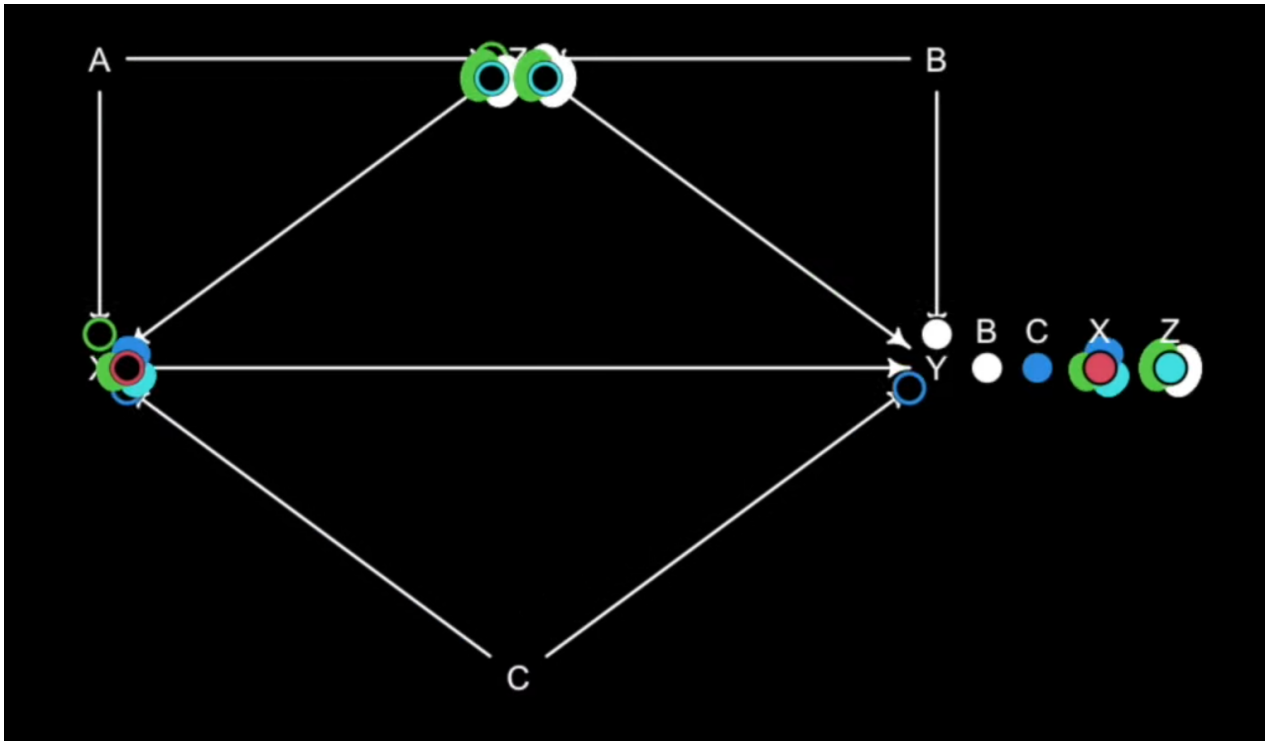


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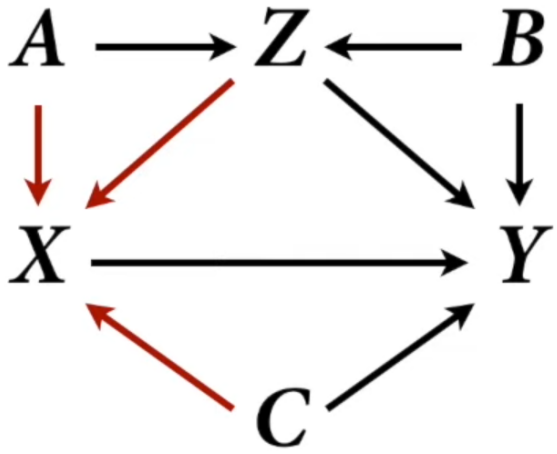


One more example

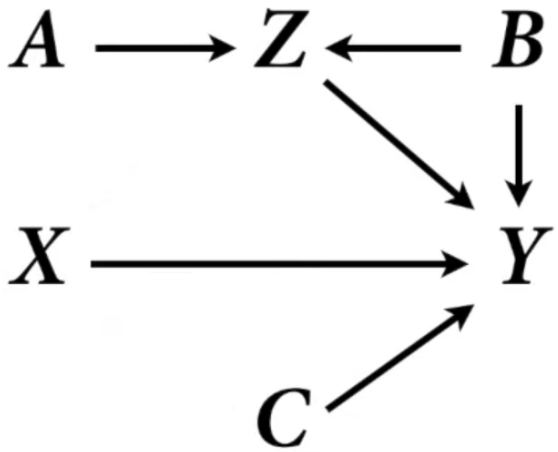


Three variables influence X by looking at its crown. Three backdoor paths?

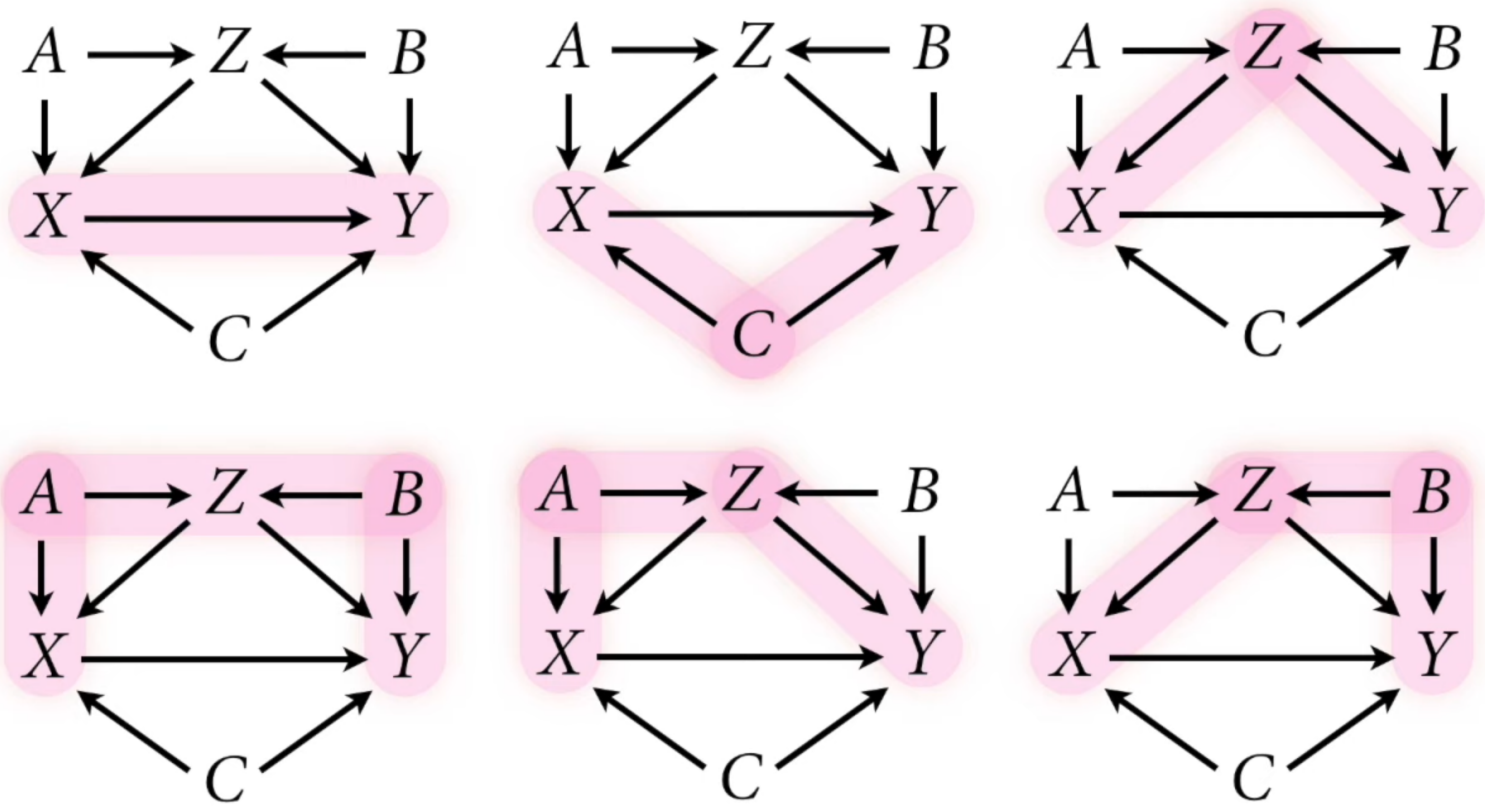
Statistically do the intervantion $\mathrm{do}(X)$



- We want $P(Y|\mathrm{do}(X))$ i.e.



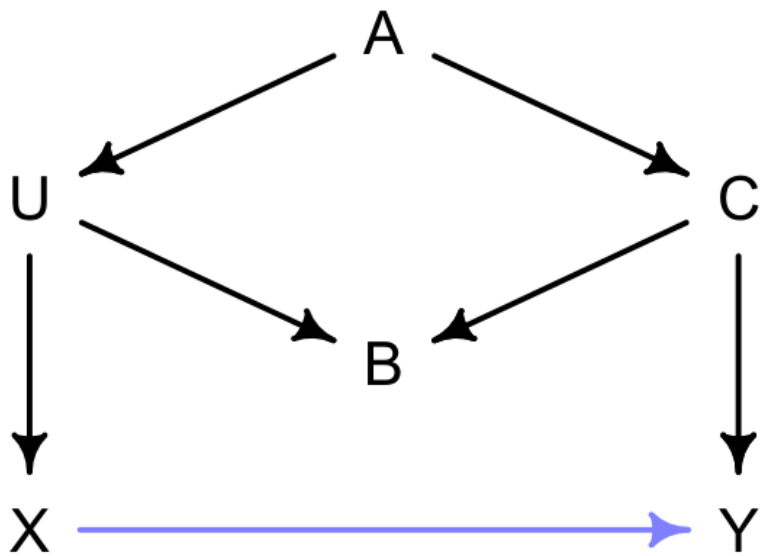
Looking at the paths one after the others



Working your way through, you should conclude that we need to condition on C, Z and either A or B. Feel free to verify that in R/`dagitty`. (also look at www.dagitty.net)

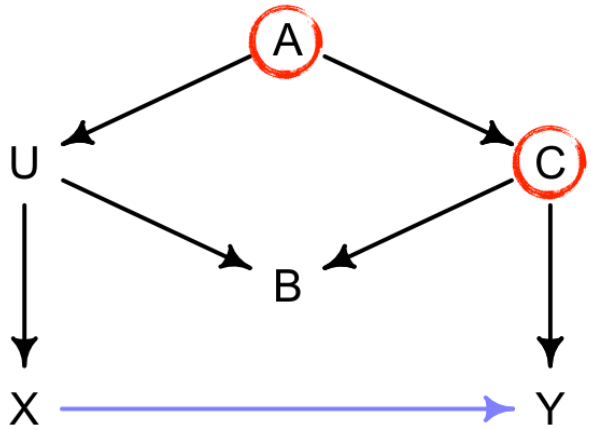
Yet another example

- Which variables, if any, should we condition on to infer the direct effect of X on Y?
- Procedure: (i) Find all paths. (ii) Open/close as necessary.



Yet another example

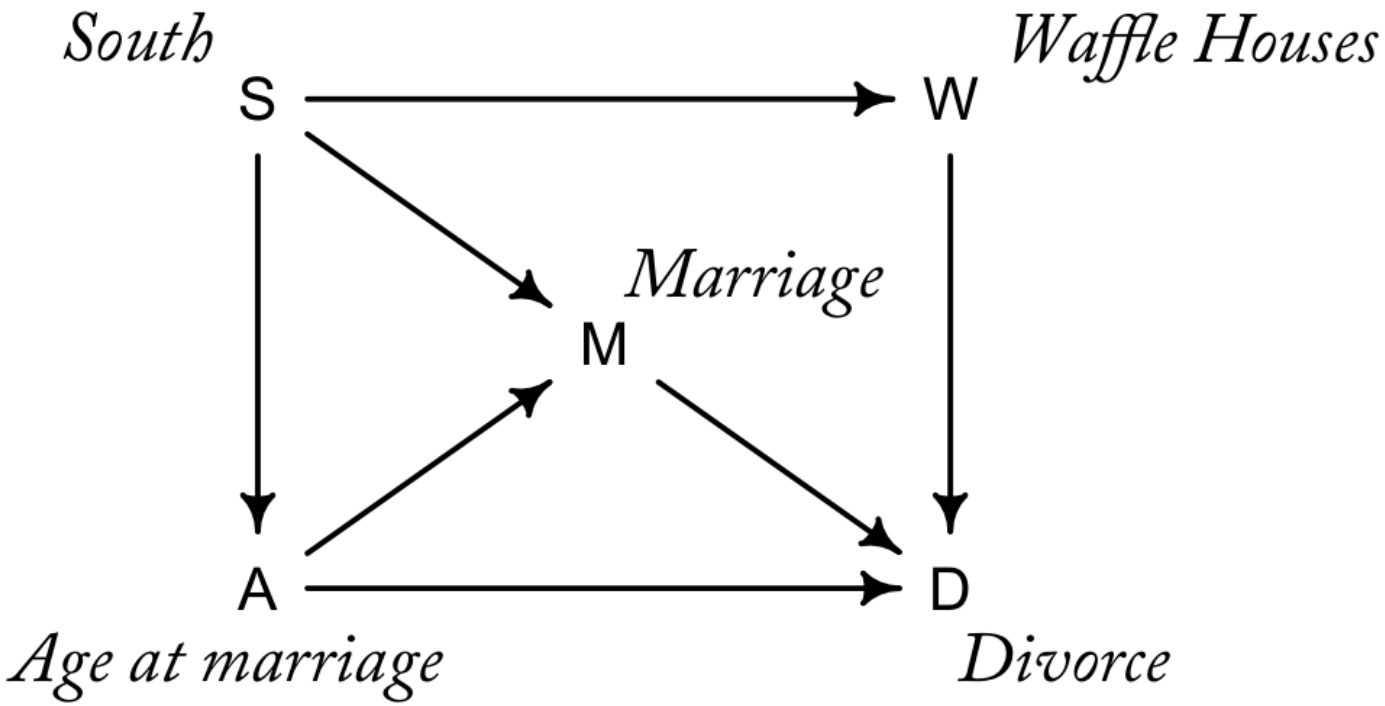
- Which variables, if any, should we condition on to infer the direct effect of X on Y?
- Procedure: (i) Find all paths. (ii) Open/close as necessary.



1. $X \rightarrow U \rightarrow A \leftarrow C \leftarrow Y$
   This path is open.
2. $X \rightarrow U \leftarrow B \rightarrow C \leftarrow Y$
   This path is closed.

Return of Waffle Houses

- Remember the Waffle House data set (marriage, divorce, etc.)?
- Which to control to infer $W \to D$?

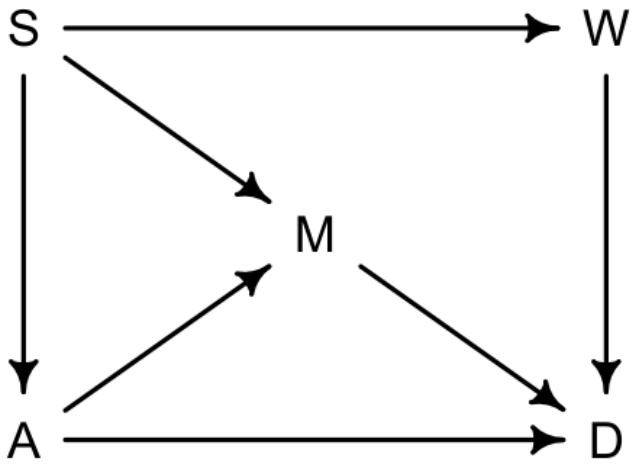


Waffle House: Implied conditional independence

- Given the DAG, we can test some implications

```
impliedConditionalIndependencies(dag_6.2)
```

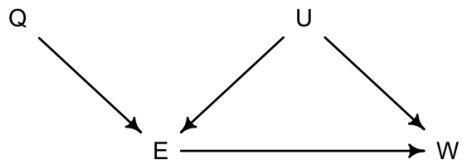
```
## A _||_ W | S
## D _||_ S | A, M, W
## M _||_ W | S
```



1. A and W are independent conditioning on S
2. D and S are independent conditioning on A, M and W.
3. M and W are independent conditioning on S.

Thoughts

- The do-calculus allowed us to extract the backdoor criterion as a rule to find adjustment sets to yield $P(Y|\mathrm{do}(X))$.
- Beware of non-causal paths that you open while closing others!
- We have seen the frontdoor criterion. Also instrumental variables.



- Notice on the way that we are saying we may not need to measure all variables in the system.

Causal inference is hard but possible

- Experiments might not always be required.
- Good, because experiments are not always practical or ethical.
  - Disease, evolution, development, dynamics of popular music, global climate, war, etc.
- Experiments must choose an intervention.
  - Interventions influence many variables at once.
  - How would you manipulate 'Obesity'?
- DAGs are small-world models. We dream of "real" models.